Koo Foundation Sun Yat-Sen Cancer Center

# A Model-Agnostic AI Agent for Automated Molecular Tumor Board Reports: A Pan-Cancer Proof-of-Concept

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## Background

- Deterministic AMP/ASCO/CAP tiering[1] takes the higher tier between OncoKB[2] and CiVIC[3].
- The rule conflates knowledge-base (KB) agreement with one-sided coverage, and with high-tier CiVIC evidence in a different tumor type, risking false Tier I calls.
- LLM variant-tiering benchmarks report over-classification. We therefore restrict LLM use to KB-discordant variants only.

| Evidence state | OncoKB | CiVIC | Rule output | |
|---|---|---|---|---|
| Both KBs agree | Level 1 | Tier I A | Tier I A | ✓ |
| One KB has no assertion | Level 1 | none | Tier I A | ✓ |
| Evidence from a different tumor | Level 3B | Tier I A* | Tier I A | ✗ |

* asserted in another tumor type: a false Tier I call

**Figure 1.** Three evidence states receive the same call under the higher-tier rule. Only the first two warrant it.

## Methods

- **Pipeline.** A `targets`-orchestrated R pipeline integrating VEP, OncoKB, CiVIC, ESCAT[4] and AMP/ASCO/CAP tiering.
- **Routing.** Concordant variants resolve deterministically. KB-discordant variants escalate to a model-agnostic agent that emits a unified tier, confidence and rationale (Table 1).
- **Benchmark.** 112 real variant–tumor cases (99 COSMIC plus 13 cross-tumor stress) against the deterministic baseline.
- **Reproducibility.** Ten cases, three runs within model and across Claude Opus 4.7, Sonnet 4.6 and Haiku 4.5 (Fleiss κ[5]).
- **Deployment.** 62 consecutive late-stage cancer patients, 8 monthly TruSight Oncology 500 batches, 11 cancer types.

| Output field | Allowed values |
|---|---|
| `unified_amp_tier` | Tier I A to Tier IV (six levels) |
| `confidence` | high · moderate · low |
| `discordance_nature` | concordant, or one of five mismatch types |
| `primary_evidence_source` | OncoKB · CiVIC · Combined · Neither |
| `resolution_rationale` | two to four sentences |

**Table 1.** JSON schema of the agent output. Every response is validated before use; the agent has no tool access.

## Architecture

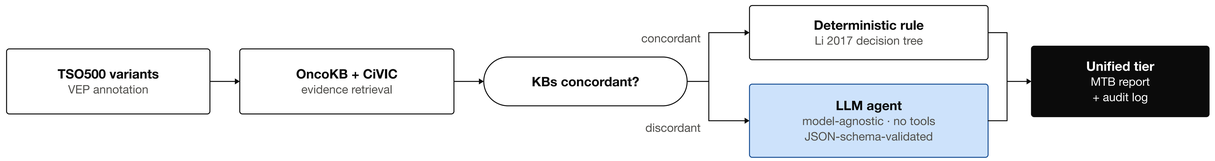



**Figure 2.** The LLM is invoked only where the two knowledge bases disagree. Every agent call is logged for audit.

**112** / 112
**Schema-valid output**
JSON schema, all cases

**80.4%**
**Tier agreement**
90 of 112 versus baseline

**0** / 70
**Tier I/II overrides**
where OncoKB is the only source

**κ 1.000**
**Within-model**
Fleiss κ, three runs

**κ 0.717**
**Cross-model**
Fleiss κ, Opus / Sonnet / Haiku

## Results: agent versus rule

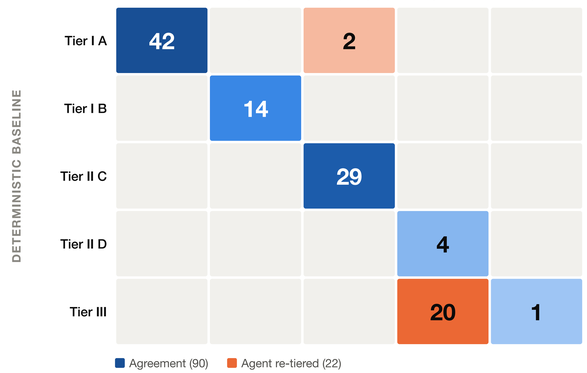



**Figure 3.** Deterministic baseline versus agent tier for 112 variant–tumor cases. Cells show case counts.

- Disagreements concentrated at documented rule blind spots.
- **20 of 112:** Tier III VUS to Tier II D, for oncogenic variants lacking a therapeutic level.
- **2 of 112:** Tier I A to Tier II C on tumor specificity.

## Reproducibility

| | | Opus 4.7 | | | Sonnet 4.6 | Haiku 4.5 |
|---|---|---|---|---|---|---|
| | | run 1 | run 2 | run 3 | | |
| KRAS G12C | NSCLC | I A | I A | I A | I A | I A |
| BRAF V600E | melanoma | I A | I A | I A | I A | I A |
| AKT1 E17K | breast | I A | I A | I A | I A | I A |
| FLT3 ITD | AML | I A | I A | I A | I A | I A |
| RET M918T | NSCLC | II C | II C | II C | II C | II C |
| ALK F1174L | NSCLC | II C | II C | II C | II C | II C |
| TP53 R175H | breast | II D | II D | II D | II D | III |
| TP53 R273H | ovarian | II D | II D | II D | II D | III |
| APC R1450* | colon | II D | II D | II D | III | III |
| GATA3 R330fs | breast | II D | II D | II D | III | III |

**Figure 4.** Tier call per case across three Opus 4.7 runs, Sonnet 4.6 and Haiku 4.5.

- Every Tier I A call and the *RET* M918T down-tier were identical across all three models.
- Cross-model variation was confined to the Tier III VUS / Tier II D boundary (Cohen κ versus Opus: Sonnet 0.722, Haiku 0.500).

## Prospective deployment

- 62 consecutive late-stage patients across 11 cancer types, in 8 monthly batches.
- All 62 cases completed end-to-end, each with an auditable per-case reconciliation log.

## Safety by evidence stratum

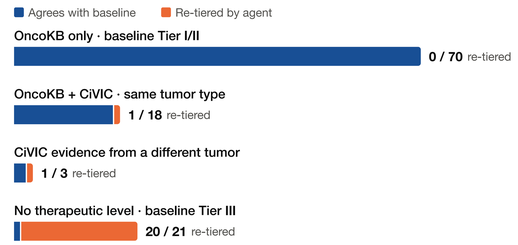



**Figure 5.** Re-tiering by the evidence available to the rule. No OncoKB-only Tier I/II call was overridden. The two Tier I A down-tiers were *RET* M918T in NSCLC and *BCOR* ITD in RCC (a histology call pending expert review).

## Case: *RET* M918T in NSCLC

| | | |
|---|---|---|
| **OncoKB** | Level 3B: approved in a different tumor type | Tier II C |
| **CiVIC** | Tier I A, but in medullary thyroid carcinoma | Tier I A |
| **Rule** | Takes the higher tier: a false Tier I call | Tier I A |
| **Agent** | Flags a tumor-specificity mismatch and follows OncoKB | Tier II C |

> "CiVIC reports TIER_I_LEVEL_A but explicitly in Medullary Thyroid Carcinoma, not NSCLC, so it cannot be applied as Tier I in the queried context."
> Agent rationale, verbatim excerpt from the audit log

## Conclusions

- Confining LLM invocation to KB-discordant variants matches deterministic reproducibility on safety-critical cases.
- The agent detects tumor-specificity errors that the rule cannot encode.
- This supports deployment as second-opinion decision support in molecular tumor boards.
- Code, prompts and schemas are open source.

**Limitations.** Agent calls were not adjudicated by an expert panel. Reproducibility was measured on a ten-case subset. Deployment was single-center, without outcome data.

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github.com/htlin222/ngs-tertiary-analysis-skills

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